

WORK EXPERIENCE

Forward Deployed Software Engineer Intern: Arboreal Management, May-November 2025, Seattle, WA

- Developed and deployed an internal tool for property management division using nodejs
- Refactored backend to be RESTful and rebuilt frontend in Svelte
- Worked with around 10 property managers in person to improve usability and add new features

TECHNICAL PROJECTS

Unreal Engine AI Editor - Assistant built in to the Unreal Engine editor that handles context and tooling to allow for AI to build projects in the editor

- AI assistant integrated into the Unreal Engine editor that can autonomously build and modify projects using in-editor tools
- Built a dynamic context pipeline that tracks user and editor state, selectively injecting relevant files, assets, and actions into LLM prompts
- Designed a tool-execution layer enabling the assistant to perform editor actions (like generating blueprints, creating scenes, and creating assets)
- Developed orchestration for back and forth tool and LLM loops with configurable round limits (this allows LLM to gather more context and perform more complicated tasks)
- Built a Slate-based UI including chat interface, model selection, attachments, and settings panels
- Used a Python vector database to handle indexing projects and assets

Tensor Trainer - Online environment for learning GPU programming (similar to LeetCode)

- Web app with virtual terminal and curriculum to teach low-level kernel programming
- Built full-stack web application, live execution environment for C++, Python, CUDA, etc
- Implemented React + TypeScript frontend with modular problem navigation, embedded code editor, and real-time terminal interface
- Developed Node.js backend with RESTful APIs to manage authentication and problem state
- Integrated Supabase (PostgreSQL) for user auth and user data storage
- Engineered per-user isolated Docker environments with resource caps provisioned dynamically when a client connects

TECHNICAL SKILLS

JavaScript, TypeScript, NodeJS, React, Python, SQL, REST apis, git, docker, C, C++, AI, end-to-end, full-stack, data models, Unix operating systems, systems programming, Svelte, agile practices

INTERESTS

Extensive travel to Ukraine, self-taught Russian: Intermediate, global politics, game development

EDUCATION

University of San Diego, Aug 2023 – May 2027 (Dean's List First Honors, Major GPA: 3.9)

- Bachelor of Science in Computer Science candidate
- Coursework: computational problem solving, programming abstractions and methods, intro to computer systems, advanced computational problem modeling, algorithms, web design